

THREE SMALL TRANSITING PLANET CANDIDATES FROM TESS AROUND TOI-678, GJ 237 AND TOI-5997

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ABSTRACT

We present three transiting planet candidates from TESS. Each was detected as a threshold-crossing event (TCE) in the multi-sector searches of the TESS pipeline but has not become a TESS Object of Interest (TOI). Around the G dwarf TOI-678, host of the candidate TOI-678.01 (11.3 d), three transits observed in 2019, 2023 and 2025 fix a period of 130.145 d and give a candidate of $3.60 \pm 0.28 R_{\oplus}$ that receives 3.9 times the Earth’s insolation. Around HIP 31126 (GJ 237), a pair of early M dwarfs 3.3’’ apart at 23.8 pc, seven transits give a 29.286-d candidate of $1.88 \pm 0.11 R_{\oplus}$ if it orbits the primary or $2.39 \pm 0.14 R_{\oplus}$ if it orbits the secondary. Around the K dwarf TOI-5997, host of the validated planet TOI-5997 b (5.65 d), nine transits give a 14.216-d candidate of $1.27 \pm 0.09 R_{\oplus}$. With a conservative allowance for correlated noise the signals have S/N of 8.4–9.8. Their depths agree from transit to transit and season to season, in simple-aperture as well as corrected photometry, and none shows an odd–even difference, a secondary eclipse or a period alias. A fit of the TESS pixel response function to difference images places each source within about 1σ of its target. The same method places a 14.8-d community TOI in the HIP 31126 field on a neighbour 33’’ away, which is most likely an eclipsing binary. With archival speckle and adaptive-optics imaging, TRICERATOPS gives false-positive probabilities of 0.021 ± 0.002 for TOI-678 and $(3.7 \pm 0.2) \times 10^{-4}$ for TOI-5997, the latter below the usual threshold for statistical validation. For HIP 31126 the value is 0.40 ± 0.02 , almost entirely because the host star is unknown: eclipsing-binary scenarios carry 0.0174 ± 0.0018 of the probability. We present the three as planet candidates and give the transit times and observations that would validate them.

Subject headings: planets and satellites: detection – techniques: photometric – binaries: visual – stars: individual: TOI-678, HIP 31126, TOI-5997 – methods: statistical

1. INTRODUCTION

The Transiting Exoplanet Survey Satellite (TESS; [Ricker et al. 2015](#)) has observed most of the sky for more than eight years. The Science Processing Operations Center (SPOC) pipeline ([Jenkins et al. 2016](#)) searches the 2-min light curves of pre-selected stars sector by sector and, every few months, in multi-sector runs that combine all sectors observed so far. A periodic signal above the detection threshold becomes a threshold-crossing event (TCE), which the Data Validation module ([Twicken et al. 2018](#); [Li et al. 2019](#)) fits and tests. TCEs that the TESS team judges planetary become TESS Objects of Interest (TOIs; [Guerrero et al. 2021](#)).

Long-period and shallow signals reach the threshold only after many sectors, so they often first appear in the latest multi-sector run, whose TCE tables are public before TOI vetting. Machine-learning classification of TCEs ([Valizadegan et al. 2025](#)), independent searches of the light curves ([Lafarga et al. 2026](#)) and uniform statistical validation of TOIs ([Collier et al. 2026](#)) show how much remains to be done with these data. Systems that already host a candidate deserve particular attention: a second transiting candidate on the same star is much less likely to be a false positive than an isolated one ([Lissauer et al. 2012](#)), and its host often already has the high-resolution imaging and spectroscopy that a validation needs.

Here we present three TCEs from these multi-sector runs. Two lie on stars that already host a planet or candidate: a 130-

d signal on TOI-678 and a 14.2-d signal on TOI-5997. The third, a 29.3-d signal, lies on HIP 31126 (GJ 237), a nearby pair of M dwarfs. None is a TOI or a community TOI (CTOI). The 14.2-d signal of TOI-5997 was listed independently as a candidate in a public GitHub repository on 2026 October 8 ([eumemic 2026](#)). The HIP 31126 signal appears in the same repository’s table of vetted signals, flagged because of a close neighbour, but is not listed as a candidate. We have not found any of the three signals in the literature.

Sections 2 and 3 describe the data and the stars, Sect. 4 the analysis, Sect. 5 the results for each system and Sect. 6 what would validate the candidates. We refer to the signals by host and period. If validated they would become TOI-678 c (or b), HIP 31126 (A or B) b and TOI-5997 c; we do not use these letters, which are reserved for validated planets.

2. DATA

2.1. TESS photometry and pipeline detections

We used the SPOC 2-min Presearch Data Conditioning simple-aperture photometry (PDCSAP; [Smith et al. 2012](#); [Stumpe et al. 2012](#)) of every sector in which each star was observed at 2-min cadence (Table 1). From the same sectors we also used the simple-aperture photometry (SAP), flux-weighted centroids, quality flags and target pixel files (TPFs). All data come from the Mikulski Archive for Space Telescopes (MAST).

All three signals are SPOC detections. The TOI-678 signal (SPOC period 130.146 d) and the HIP 31126 signal (29.286 d)

are TCEs of the multi-sector search of Sectors 1–96, with pipeline signal-to-noise ratios (S/N) of 9.6 and 10.8 and multiple-event statistics of 8.7 and 9.5. For TOI-678 that run contains all three transits; the first two had already been detected in the runs of Sectors 1–65 and 1–69 as a two-transit TCE, with a period equal to their 1431.6-d spacing. For HIP 31126 it contains five of the seven transits used here; the other two are in Sector 98. The 14.216-d signal of TOI-5997 is a TCE of the search of Sectors 14–86 (S/N 11.3, multiple-event statistic 8.9). None of the three has become a TOI. For the HIP 31126 TCE, the Data Validation difference-image centroid lies $5.2 \pm 4.3''$ from the catalogue position of the brighter star, HIP 31126 A; for the TOI-678 TCE, Data Validation gave no centroid result.

2.2. High-resolution imaging

All three stars have archival high-resolution imaging on the Exoplanet Follow-up Observing Program website (ExoFOP). HIP 31126 was observed with the speckle camera HRCam on the 4.1-m SOAR telescope in the I band on 2019 May 18 (Tokovinin 2018; Ziegler et al. 2020). The image resolves the companion B at $3.26''$ (position angle 219° , $\Delta I = 1.5$) and shows no other source; the 5σ limits relative to A, which we digitised from the sensitivity curve posted with the data, are $\Delta I = 2.5$ at $0.15''$ and 5.4 at $1''$. TOI-678 was observed with the Zorro speckle imager on Gemini-South at 562 and 832 nm on 2020 December 28 and 2023 March 4 (Howell et al. 2011; Scott et al. 2021); we use the 2020 curves, the only ones posted as tables, which reach $\Delta m_{832} = 4.8$ at $0.1''$ and 6.6 at $1''$. TOI-5997 was observed with ‘Alopec on Gemini-North at 562 and 832 nm on 2024 May 23 ($\Delta m_{832} = 7.2$ at $0.5''$) and with PHARO (Hayward et al. 2001) on the Palomar 5-m telescope in B_{r} on 2023 June 6. No source other than HIP 31126 B was reported near any of the targets. An observation of TOI-5997 with the SAI 2.5-m speckle polarimeter is also on ExoFOP, without a tabulated contrast curve; we do not use it.

2.3. Spectroscopy and astrometry

ExoFOP lists stellar parameters from CHIRON (Tokovinin et al. 2013) spectra of TOI-678 taken at two epochs in 2019 and from two TRES spectra of TOI-5997 taken in 2023, analysed with the Stellar Parameter Classification code (SPC; Buchhave et al. 2012); TOI-678 also has two CORALIE (Queloz et al. 2000) spectra from 2019. No radial velocities from these spectra are public. Gaia DR3 (Gaia Collaboration et al. 2016, 2023) measured the radial velocities of TOI-678, TOI-5997 and HIP 31126 B over 12–20 epochs (Katz et al. 2023) and found no variability (probabilities of constancy $p = 0.28, 0.10$ and 0.92). All three targets and HIP 31126 B have renormalised unit weight errors (RUWE; Lindgren et al. 2021) between 0.90 and 1.03 and no non-single-star solution. We also use 2MASS photometry (Skrutskie et al. 2006).

3. THE STARS

Table 2 lists the stellar parameters.

For TOI-678 and TOI-5997 we computed the luminosity from the Gaia G magnitude, the parallax and the bolometric correction of Andrae et al. (2018), with $E(B - V) = 0.045$ for TOI-678 and none for TOI-5997. We took T_{eff} , $\log g$ and $[M/H]$ from the spectra, with T_{eff} uncertainties of at least 2% (Tayar et al. 2022). We derived the radius from the Stefan–Boltzmann law and the mass from the relations of Torres et al.

(2010), which take the spectroscopic T_{eff} , $\log g$ and $[M/H]$ as input. The $\log g$ implied by mass and radius agrees with the spectroscopic values (4.34 and 4.62). TOI-678 is a slightly metal-rich G7–G8 dwarf at 207 pc; TOI-5997 is a metal-poor K3–K4 dwarf at 46.6 pc.

The proper motions of HIP 31126 A and B differ by 21.7 mas yr^{-1} , or 2.5 km s^{-1} at their distance. This is less than the orbital speed of 3.3 km s^{-1} for a circular orbit at their projected separation of 79 au ($3.33''$ at the Gaia epoch 2016.0). Their parallaxes differ by $0.072 \pm 0.019 \text{ mas}$, formally 3.9σ , but Gaia parallax uncertainties are underestimated for pairs this close (El-Badry et al. 2021). The pair is consistent with being bound, and we treat it so. Its 2MASS photometry is partly blended: B has a poor-quality K_s measurement, and A’s catalogued K_s is 0.22 mag brighter than its Gaia photometry predicts, as expected if it includes some of B’s light. We therefore derived both stars’ parameters from Gaia. We converted the absolute magnitude M_G into M_{K_s} and T_{eff} with the dwarf sequence of Pecaut & Mamajek (2013), in its updated online version¹ (uncertainties 0.1 mag and 100 K), and then obtained the radius from the radius– M_{K_s} relation of Mann et al. (2015), the mass from the mass– M_{K_s} relation of Mann et al. (2019), and the luminosity from radius and T_{eff} . Using A’s catalogued K_s instead would give a radius 7% larger. A and B are early M dwarfs, about M0 and M2.5.

The TESS Input Catalog lists B with $T = 9.53$, a magnitude brighter than its Gaia photometry implies ($T = 10.55$ from the relations of Stassun et al. 2019), probably because the catalogue value comes from the blended 2MASS photometry. The SPOC crowding correction used the catalogue value, so the PDCSAP light curve of A over-corrects for B’s light. We computed the fraction of the flux in the SPOC aperture that comes from each star with the TESS pixel response function (PRF) and the registered positions of all catalogued stars (Sect. 4.3): A contributes 0.76 and B 0.21, against the SPOC estimate of 0.592 for A. A transit on A therefore appears in PDCSAP with $g_A = 1.287$ times its true depth in A’s light, and a transit on B with $g_B = 0.350$ times its depth in B’s light. For TOI-678 and TOI-5997 our flux fractions agree with the SPOC values to within 2%.

4. ANALYSIS

The same code analysed all three systems. Four other signals in the same light curves serve as controls: TOI-5997 b, validated by Collier et al. (2026); TOI-678.01; a 6.56-d TCE of TOI-678; and a 14.8-d signal in the HIP 31126 light curve that is listed as the CTOI TIC 293689267.01 (Appendix A). Before looking at the light curves, we fixed the criteria that a signal had to meet. The main one was $S/N \geq 10$, with correlated noise estimated by the time-averaging method of Sect. 4.2.

4.1. Light curves

We removed cadences with non-zero quality flags and detrended each sector with a time-windowed biweight filter (Hippke et al. 2019) of width $\max(0.75 \text{ d}, 3T_{14})$, where T_{14} is the transit duration, with the transits of all known signals masked. Upward outliers more than five times the robust scatter above the trend were removed. A second detrending, with the same window and no masks, was used for blind searches.

¹ Version 2022.04.16 of https://www.pas.rochester.edu/~emamajek/EEM_dwarf_UBVIJHK_colors_Teff.txt.

TABLE 1
TESS 2-MIN DATA USED.

Star	Sectors	Transits used	CROWDSAP	2-min scatter (ppm)
TOI-678	7, 8, 9, 30, 33, 34, 35, 36, 61, 62, 67, 69, 87, 88, 89, 90, 94, 96	3 (S8, S61, S90)	0.892	1662
HIP 31126	6, 7, 33, 34, 61, 87, 88, 98	7	0.592	1066
TOI-5997	25, 26, 52, 53, 79, 80	9 (+1)	0.990	713

CROWDSAP is the SPOC estimate of the fraction of the flux in the aperture that comes from the target, averaged over sectors. The scatter is 1.4826 times the median absolute deviation of the detrended 2-min flux. Transits used are those with at least 60% of the in-transit and of the surrounding out-of-transit cadences. HIP 31126 has an eighth, partial transit at the end of Sector 6, and TOI-5997 a tenth that overlaps a transit of TOI-5997 b (Sect. 5.3).

TABLE 2
STELLAR PARAMETERS.

	TOI-678	HIP 31126 A	HIP 31126 B	TOI-5997
Other names	TIC 294395926	GJ 237, TIC 293689266	TIC 293689267	HIP 85850, TIC 39516274
Gaia DR3	5490097505810081024	5568872394338114560	5568872398635858816	4602034956332491264
Parallax (mas)	4.831 ± 0.013	41.991 ± 0.012	42.063 ± 0.015	21.450 ± 0.012
Distance (pc)	206.98 ± 0.55	23.81 ± 0.01	23.77 ± 0.01	46.62 ± 0.03
$\mu_{\alpha*}, \mu_{\delta}$ (mas yr ⁻¹)	$-31.33, +71.64$	$-277.53, -29.76$	$-263.67, -13.04$	$-223.52, +177.32$
RUWE	0.90	1.01	0.98	1.03
Gaia RV (km s ⁻¹)	$+40.1 \pm 0.4$	—	$+11.0 \pm 0.3$	-27.2 ± 0.2
$G; G_{BP} - G_{RP}$ (mag)	$11.40; 0.92$	$10.02; 1.91$	$11.60; 2.42$	$9.90; 1.26$
T (mag)	10.92	9.13^a	10.55^a	9.26
$J; K_s$ (mag)	$10.26; 9.85$	$7.63; 6.79^b$	blended	$8.32; 7.68$
T_{eff} (K)	5536 ± 148	3864 ± 99	3465 ± 99	4717 ± 106
$\log g$ (cgs)	4.38 ± 0.08	4.66 ± 0.03	4.82 ± 0.03	4.60 ± 0.08
[M/H]	0.10 ± 0.08	—	—	-0.34 ± 0.08
R_* ($R_{\odot}$)	1.088 ± 0.066	0.588 ± 0.025	0.411 ± 0.018	0.687 ± 0.042
M_* ($M_{\odot}$)	1.036 ± 0.086	0.578 ± 0.019	0.404 ± 0.018	0.686 ± 0.053
ρ_* (g cm ⁻³)	1.13 ± 0.27	4.00 ± 0.44	8.19 ± 0.90	2.98 ± 0.69
L_* ($L_{\odot}$)	1.00 ± 0.02	0.069 ± 0.009	0.0219 ± 0.0032	0.211 ± 0.007

Astrometry and G photometry from Gaia DR3; J, K_s from 2MASS; T_{eff} and [M/H] of TOI-678 and TOI-5997 from the CHIRON and TRES spectra on ExoFOP; other parameters derived as described in Sect. 3 ($\log g$ from M_* and R_*). ^aFrom the Gaia colours (Stassun et al. 2019); the TIC lists 9.06 and 9.53.

^bProbably includes some light from B.

4.2. Tests of each signal

The tests use a box-shaped measurement of the depth: the mean flux in the central 80% of the transit compared with the flux 1–3 transit durations on either side. They use the ephemerides and durations of preliminary fits, which differ from the final values (Table 4) by at most 20 min in mid-transit time (for the earliest TOI-678 transit) and 6% in duration; this changes the box depths by a few per cent at most. We measured this depth at the candidate’s ephemeris for all transits together, per transit, per observing season, for odd and even transits, and in SAP flux. We also measured it at phase 0.5, at every other phase of the orbit, and at the extra epochs that periods of $P/2$ to $P/10$ would imply. We also searched each light curve with a box least-squares (BLS) periodogram (Kovács et al. 2002) up to a maximum period set for each star (Table 3), with trial durations of 0.6–1.3 times the candidate’s and with the other known signals masked. Finally, we measured the shift of the flux-weighted centroid in transit.

Correlated noise enters through a factor β by which the white-noise uncertainties are multiplied. We estimated it in two ways. The full-phase β is the scatter of the depth S/N measured at every phase of the orbit away from the transit. The time-averaging β compares the scatter of the out-of-transit residuals binned to the transit duration with the white-noise expectation (Pont et al. 2006; Winn et al. 2008); it is floored at 1. The full-phase value is the larger, and Table 3 gives the S/N and the consistency tests with both. As a check that does not depend on a noise model, we compared the depth at the candidate’s ephemeris with the depths at 3000 randomly chosen phases of the same period; because the boxes overlap, these correspond to only P/T_{14} (110–490) independent trials.

4.3. Pixel-level localisation

The flux-weighted centroid is a blunt tool in crowded TESS apertures. To find where each signal comes from, we fitted the TESS PRF to difference images, as Bryson et al. (2013) did for Kepler; Higgins & Bell (2023) use a related PRF fit to locate variable stars in TESS data. For every transit with full coverage in a 2-min TPF we subtracted the mean image in transit (within $0.4 T_{14}$ of mid-transit) from the mean of the images 0.75 – $1.75 T_{14}$ before and after it, and averaged the transits of each sector. The same was done at 60 random epochs that avoid all known signals, giving null difference images from which we estimated the per-pixel noise.

The PRF of each sector came from the TESS PRF models on MAST, interpolated to the target’s position on the detector. The catalogue positions of all stars in the TIC within 0.07° , moved by their proper motions to the sector epoch, were registered to the mean image by fitting all stars at once with a free common shift. We then fitted all sectors’ difference images together with a single point source at a free sky position, with a non-negative flux decrement proportional to the target’s registered flux in each sector and a free constant per sector.

We tested each star that could produce the signal (one whose eclipse would need to be shallower than 100% to give the observed depth in the aperture) in two ways. The first uses the covariance of 500 bootstrap fits, in which the best-fitting model was added to averages of randomly chosen null images and refitted from a random start within $6''$. The distance of the star from the fitted position, measured in units of this covariance (the Mahalanobis distance), is converted to a Gaussian significance. The second uses the increase in χ^2 when the source is placed on the star, divided by a factor that

calibrates the per-pixel χ^2 for correlated noise. This factor is the median, over the bootstrap fits, of the χ^2 at the true position minus the minimum χ^2 , divided by the median of a χ^2 distribution with two degrees of freedom. It is 3.7–4.4 for the three candidates (2.4–4.6 including the control signals), so per-pixel errors alone would overstate the significance of an exclusion. We regard a star as excluded only if both tests put it at 3σ or more.

4.4. Transit fits

We fitted each light curve with *batman* (Kreidberg 2015) models sampled with *emcee* (Foreman-Mackey et al. 2013), using all data within three transit durations of each transit, on circular orbits. For TOI-678 and TOI-5997 the candidate and TOI-678.01 or TOI-5997 b were fitted together; other signals were masked. Free parameters were, for each planet, the period, the reference epoch, $R_p/R_\star$ and the impact parameter; the stellar density, with a Gaussian prior from Table 2; the quadratic limb-darkening coefficients in the parametrisation of Kipping (2013), with Gaussian priors of width 0.15 centred on approximate values for the TESS band from Claret (2017); and a white-noise scale factor per season. For HIP 31126 the model transit is multiplied by g_A or g_B , and the fit was run once for each host. We used 48 walkers and 15 000 steps and discarded the first 5000. The chains are 23–63 autocorrelation times long; the shortest, for TOI-678, still gives about 1105 independent samples, enough for the medians and 68% intervals we quote. The fits treat the noise as white within each season, so their uncertainties on period and epoch may be too small by up to the factor β ; we inflate the timing uncertainties of the predicted transits accordingly. To check that the transit shape agrees with the star, we refitted each candidate alone with $a/R_\star$ free and compared the implied stellar density with Table 2; these shorter chains (8000 steps, 13–16 autocorrelation times) serve only as a check.

4.5. False-positive probabilities

We used TRICERATOPS 1.1.0 (Giacalone et al. 2021, 2022) to compute false-positive probabilities. The inputs were the phase-folded light curve within two transit durations of mid-transit (other planets divided out, 200 bins); the TIC stars within 0.07° ; the SPOC apertures of the sectors with transits; and the contrast curves of Sect. 2.2. We placed the TIC stars at their positions at the mean epoch of the observations, kept one entry for every Gaia source, and used our parameters for the targets and for HIP 31126 B; TRICERATOPS assigns solar values to the few faint neighbours without catalogued properties. The SOAR *I*-band and 832-nm curves were treated as TESS band, the 562-nm curves as the visible band and the Bry curve as *K*; TRICERATOPS adopts the tightest constraint. We replaced each contrast curve, where needed, by its lower envelope (the smallest contrast reached at any larger separation) so that it increases with separation. For the blended-background scenarios we supplied the Gaia DR3 stars within 0.1 deg^2 of each target, which TRICERATOPS accepts in place of a synthetic star field from the TRILEGAL model. Because PDCSAP over-corrects HIP 31126 for B’s light, we rescaled its light curve to the depth seen in the aperture before passing it to TRICERATOPS.

TRICERATOPS reports the false-positive probability (FPP), the probability that the signal is not from a planet transiting the target, and the nearby false-positive probability (NFPP), the part of it due to resolved neighbours. Our

adopted runs use the imaging but not the pixel analysis, because the calibration of its significances has not been tested on a large sample. We ran them four or five times for each candidate with 10^6 draws and different random seeds, and report the mean and standard deviation. We also ran a version in which the stars excluded by the pixel analysis cannot host the signal, and one without imaging.

5. RESULTS

Figure 1 shows the transits and Table 3 the tests. The fitted parameters are in Table 4, the pixel analysis in Fig. 2 and the false-positive probabilities in Table 5.

5.1. TOI-678: a 130-day sub-Neptune candidate

TOI-678 shows three transits, in Sectors 8 (2019 February), 61 (2023 January) and 90 (2025 March), with depths of 982 ± 161 , 942 ± 149 and 640 ± 158 ppm. The third is shallower, but the depths are consistent ($p = 0.25$, or 0.16 with the time-averaging β). TOI-678.01 is also shallower in that season (Sectors 87–90): 671 ± 73 ppm, or 21% below its mean, which points to an instrumental effect common to those sectors. The intervals between the transits are 11 and 6 times 130.145 d. Because the true period must divide both intervals, and 11 and 6 have no common factor, it can only be 130.145 d or an integer fraction of it. The 18 sectors of 2-min data rule out every fraction from $P/2$ to $P/10$ (Table 3), and shorter periods would have produced many more transits in the data. The transits have the same depth in SAP flux and show no odd–even difference and no dip at phase 0.5. The deepest dip elsewhere in the phase scan (3.2σ , at phase 0.42) is consistent with chance, given about 488 independent trials.

Numbering the epochs from the Sector 61 transit, three other predicted transits fall close to TESS data. Epoch +7 came 1.2 d after the end of the Sector 94 data, and epoch –12 4.3 d after the end of the Sector 3 full-frame images; TOI-678 was not observed in Sector 4. Epoch –10 (BTJD 1670.80, where BTJD = BJD_{TDB} – 2457000) lies in Sector 13, observed only in full-frame images, where the 30-min TESS-SPOC light curve (Caldwell et al. 2020) shows variations of 1–2 ppt over a day, larger than the transit. A fit with the transit shape fixed and a quadratic baseline gives a depth of $1.10^{+0.35}_{-0.35}$ times the expected one in the TESS-SPOC PDCSAP flux, $0.98^{+0.27}_{-0.27}$ in its SAP flux and $0.45^{+0.25}_{-0.25}$ in the QLP light curve (Huang et al. 2020a,b), so these data neither confirm nor exclude the transit. In Sector 6, at an epoch where a period of $P/2$ would place a transit, the TESS-SPOC light curve shows none (depth $-0.15^{+0.15}_{-0.18}$ times the expected).

Our blind search of all 18 sectors does not put the 130-d signal at the top: with only three transits, several peaks of similar power appear between 69 and 138 d, and the unmasked 0.8-d biweight filter used for the blind search removes part of a 6-h transit. Without the SPOC detection our search would not have found it. At its ephemeris the signal is deeper than at any of the 3000 random phases (8.5σ).

The pixel analysis places the source 0.3σ from TOI-678. Once the correlated pixel noise is calibrated, however, the transit is detected in the difference images at only 2.7σ . The calibrated χ^2 test therefore cannot exclude any star at 3σ , and no neighbour is excluded by both tests. The closest neighbours, TIC 766625715 ($T = 16.5$, $19''$) and TIC 294395924 ($T = 15.7$, $20''$), are 3.7σ and 3.9σ from the fitted position by the bootstrap test but only 1.8σ and 1.9σ by the calibrated χ^2 ; they would need eclipses of 18% and 9% of their light.

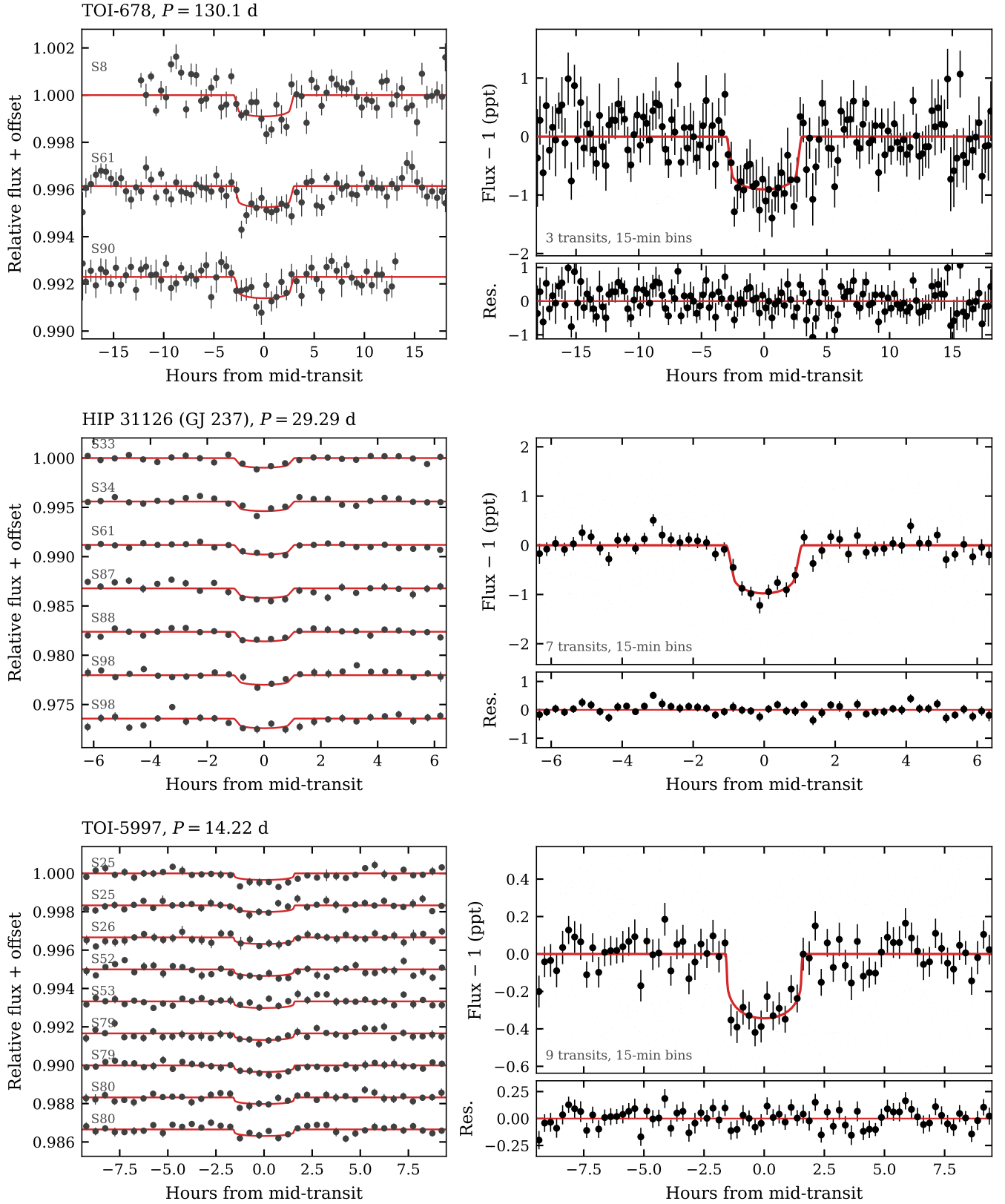


FIG. 1.— TESS 2-min photometry of the three candidates. Left: each transit used, in 30-min bins, offset vertically and labelled with its sector. Right: all of them phase-folded, with 2-min data (grey), 15-min bins (black) and the adopted model (red; for HIP 31126, the model for host A including the factor g_A), and the residuals.

TABLE 3

TESTS OF THE CANDIDATES. DEPTHS ARE BOX DEPTHS IN PDCSAP FLUX, IN PPM; THEIR UNCERTAINTIES INCLUDE THE FULL-PHASE β .

Test	TOI-678 (130.1 d)	HIP 31126 (29.29 d)	TOI-5997 (14.22 d)
Transits used	3	7	9
Box depth	869 $\pm$ 90	967 $\pm$ 99	330 $\pm$ 39
β : full phase; time averaging	1.15; 1.00	1.63; 1.45	1.42; 1.10
S/N: full-phase β ; time-averaging β	9.7; 11.1	9.8; 11.0	8.4; 10.8
Depth at random phases: significance; deeper	8.5 σ ; 0 of 3 000	9.7 σ ; 0 of 3 000	7.5 σ ; 0 of 3 000
Per-transit depths, p (χ^2/dof): full phase; time averaging	0.25 (2.8/2); 0.16	0.89 (2.3/6); 0.82	0.34 (9.1/8); 0.06
Per-season depths, p : full phase; time averaging	one transit per season	0.79; 0.73	0.49; 0.30
Odd; even depth	982 $\pm$ 161; 800 $\pm$ 108	914 $\pm$ 145; 1004 $\pm$ 136	292 $\pm$ 60; 365 $\pm$ 51
Odd–even difference (σ)	0.9	0.5	0.9
Depth at phase 0.5 ^a	−209 $\pm$ 96	−216 $\pm$ 90	41 $\pm$ 50
Deepest dip elsewhere in phase (σ)	3.2	2.6	2.9
SAP depth; divided by CROWDSAP	719 $\pm$ 84; 806 $\pm$ 94	527 $\pm$ 59; 889 $\pm$ 99	349 $\pm$ 43; 353 $\pm$ 43
Largest dip at the $P/2$ to $P/10$ epochs (σ)	0.7	0.6	0.8
Blind search: rank (period range searched)	not in top 5 (1–150 d) ^b	1 (1–60 d)	1 (1–40 d)
Centroid shift, column; row (σ)	−0.1; 0.1	0.2; 0.3	0.6; −0.9
Pixel-level source position ($''$, E; N)	−0.8, +2.9 ($\pm$ 6.9, 4.6)	−1.4, −4.5 ($\pm$ 3.3, 3.5)	+0.0, −5.4 ($\pm$ 6.0, 5.5)
Target from fitted position: Mahalanobis; χ^2 (σ)	0.3; 0.2	A: 0.8; 1.0 B: 0.2; 0.4	0.5; 0.7
Neighbours able to cause the signal; excluded	9; 0	5 ^c ; 4	4; 0

^aThe epochs at phase 0.5 are also the extra epochs of a period $P/2$. ^bSee Sect. 5.1. ^cIncluding B, which cannot be excluded.

TABLE 4

TRANSIT FITS: POSTERIOR MEDIANS WITH 68% INTERVALS.

	TOI-678	HIP 31126 (host A)	HIP 31126 (host B)	TOI-5997
P (d)	130.1455 $\pm$ 0.0007	29.28559 $\pm$ 0.00006	29.28558 $\pm$ 0.00006	14.21604 $\pm$ 0.00005
T_0 (BJD _{TDB} − 2457000)	2972.2556 $\pm$ 0.0048	3334.9927 $\pm$ 0.0018	3334.9926 $\pm$ 0.0016	2758.2834 ^{+0.0022} _{−0.0032}
$R_p/R_\star$	0.0304 $\pm$ 0.0016	0.0293 $\pm$ 0.0013	0.0533 $\pm$ 0.0021	0.0170 $\pm$ 0.0007
$a/R_\star$	104.7 ^{+2.4} _{−4.3}	56.6 $\pm$ 2.1	72.3 $\pm$ 2.5	32.9 $\pm$ 1.9
Impact parameter b	0.81 $\pm$ 0.02	0.88 $\pm$ 0.02	0.79 $\pm$ 0.03	0.33 $\pm$ 0.20
T_{14} (h)	6.05 $\pm$ 0.21	2.14 $\pm$ 0.07	2.17 $\pm$ 0.07	3.16 $\pm$ 0.12
$(R_p/R_\star)^2$ (ppm)	927 $\pm$ 95	856 $\pm$ 75	2841 $\pm$ 222	289 $\pm$ 24
R_p ($R_\oplus$)	3.60 $\pm$ 0.28	1.88 $\pm$ 0.11	2.39 $\pm$ 0.14	1.27 $\pm$ 0.09
a (au)	0.509 $\pm$ 0.014	0.155 $\pm$ 0.002	0.137 $\pm$ 0.002	0.101 $\pm$ 0.003
Insolation ($S_\oplus$)	3.9 $\pm$ 0.2	2.9 $\pm$ 0.4	1.16 $\pm$ 0.17	21 $\pm$ 1
T_{eq} (K) ^a	391 $\pm$ 6	363 $\pm$ 12	289 $\pm$ 11	593 $\pm$ 9
$\rho_\star$ from the fit with $a/R_\star$ free (g cm^{-3})	3.8 ^{+2.1} _{−2.6}	17.8 ^{+14.0} _{−13.2}	—	2.7 ^{+1.3} _{−1.9}
Predicted mass ($M_\oplus$) ^b	12.7	4.2	6.3	2.2
Predicted RV semi-amplitude (m s^{-1})	1.6	1.3	2.4	0.7

Fitted jointly with the candidates: TOI-678.01, $P = 11.329893 \pm 0.000013$ d, $R_p = 3.33 \pm 0.21 R_\oplus$; TOI-5997 b, $P = 5.654997 \pm 0.000012$ d, $R_p = 1.44 \pm 0.11 R_\oplus$ ($1.45 \pm 0.14 R_\oplus$ in Collier et al. 2026). Insolation and T_{eq} use a from Kepler’s third law. ^aZero albedo, full heat redistribution. ^bFrom the mass–radius relation of Chen & Kipping (2017).

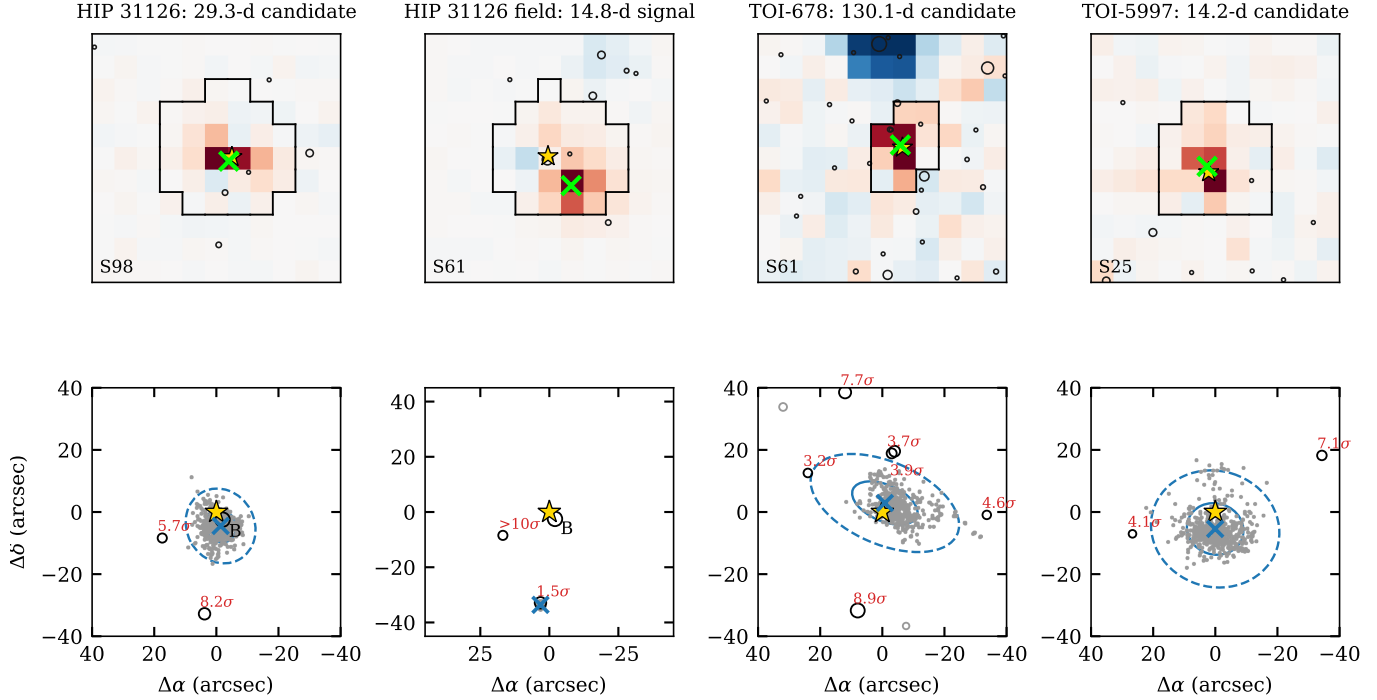


FIG. 2.— Pixel-level localisation. Top: mean difference image (out of transit minus in transit) of the sector with the clearest signal, in the TESS pixel frame (not north-up), with the SPOC aperture (black outline), the target (star), other catalogued stars (circles, larger for brighter stars) and the fitted source position (green cross). Bottom: the fitted source position on the sky relative to the target (blue cross) with 1σ and 3σ contours from the bootstrap (grey dots), and the stars able to produce the signal, labelled with their Mahalanobis distance from the fitted position (grey circles: stars too faint to produce it). The second column spans $\pm 45''$, the others $\pm 40''$. The second column is a control: the 14.8-d signal of the CTOI TIC 293689267.01, which the fit places on a $T = 13.4$ star $33''$ south of HIP 31126.

The transit duration is consistent with a planet around TOI-678: the free fit gives $\rho_\star = 3.8^{+2.1}_{-2.6} \text{ g cm}^{-3}$, 1.0σ from the value in Table 2. The candidate has $R_p = 3.60 \pm 0.28 R_\oplus$ on a 0.509 ± 0.014 -au orbit, where it receives $3.9 \pm 0.2 S_\oplus$ and has an equilibrium temperature of 391 ± 6 K.

TOI-678.01, fitted jointly, has $R_p = 3.33 \pm 0.21 R_\oplus$, consistent with its TOI value of $3.4 \pm 0.3 R_\oplus$. The 6.56-d TCE reaches only S/N 8.7 with the time-averaging β , below our threshold of 10, and we do not consider it here (Appendix A).

TRICERATOPS gives FPP = 0.021 ± 0.002 and NFPP = 0.016 ± 0.002 (Table 5). Most of the NFPP (0.013 ± 0.002) comes from TIC 294395934 ($T = 12.8$, $33''$), which would need an eclipse of only 1.2%; the bootstrap test puts it 8.9σ from the fitted position, the calibrated χ^2 test only 2.7σ . Without the contrast curves the FPP is 0.025.

5.2. HIP 31126 (GJ 237): a small planet candidate around one of two M dwarfs

Seven transits of HIP 31126 in six sectors between 2021 January and 2025 December have consistent depths ($p = 0.89$ per transit and 0.79 per season; 0.82 and 0.73 with the time-averaging β), no odd–even difference and no secondary eclipse. A blind search ranks the 29.29-d period first, with a BLS S/N of 17.5 against 14.1 for the next peak. The SAP depth divided by the SPOC flux fraction, 889 ± 99 ppm, agrees with the PDCSAP depth. The signal is not a half-period alias: at the extra epochs of $P/2$ the flux is slightly higher (-216 ± 90 ppm), not lower.

The difficulty is the host. A and B are $3.3''$ apart, a sixth of a TESS pixel, and both lie inside the aperture. The fitted source position is 0.8σ (bootstrap) and 1.0σ (calibrated χ^2) from A,

and 0.2σ and 0.4σ from B, so the pixel analysis cannot tell them apart. All 4 other stars able to produce the signal are excluded; the nearest (TIC 293689268, $19''$) is at 5.7σ and 3.6σ . Nor does the transit shape separate them. With A's density the fit needs an impact parameter $b = 0.88 \pm 0.02$, and with B's density $b = 0.79 \pm 0.03$. The fit with free density gives $\rho_\star = 17.8^{+14.0}_{-13.2} \text{ g cm}^{-3}$, 1.0σ from A and 0.7σ from B. The candidate would have $R_p = 1.88 \pm 0.11 R_\oplus$ and receive $2.9 \pm 0.4 S_\oplus$ if it orbits A, and $R_p = 2.39 \pm 0.14 R_\oplus$ and $1.16 \pm 0.17 S_\oplus$ if it orbits B.

TRICERATOPS reflects this. A planet on A carries 0.60 ± 0.02 of the probability and a planet on B 0.38 ± 0.02 . Eclipsing-binary scenarios together carry 0.0174 ± 0.0018 , mostly on two neighbours that the pixel analysis excludes; with those exclusions they carry 4.8×10^{-4} . The FPP of 0.40 ± 0.02 therefore mostly reflects the unknown host: it is essentially the probability that the candidate does not orbit A. The probability that the signal is a planet in the HIP 31126 system is 0.9826 ± 0.0018 .

The pixel analysis also locates the source of the 14.8-d signal in the same light curve, a strong SPOC TCE (S/N 32) that was submitted as a CTOI on B by R. Cloutier in 2019 June using the ORION pipeline (Cloutier 2019). Its source is 1.5σ from TIC 293689273, a $T = 13.4$ star $33''$ south of A, and more than 10σ from A and B (Fig. 2). That star would need a 5.8% eclipse. The signal also has an odd–even difference of 2.4σ in our reduction, and the SPOC fit gives it a 5.5-h duration, longer than a planet could transit A or B at this period (at most 3.3 and 2.6 h). The CTOI is most likely an eclipsing binary on that neighbour, and it does not affect the 29.29-d

candidate.

5.3. TOI-5997: a second small planet candidate

TOI-5997 shows nine transits of the 14.216-d signal in three seasons, with consistent depths ($p = 0.34$ per transit, or 0.06 with the time-averaging β ; $p = 0.49$ per season), no odd–even difference, no dip at phase 0.5, and the same depth in SAP flux. A tenth transit, at BTJD 2019.05, overlaps a transit of TOI-5997 b whose mid-time is 1.0 h later; it is masked in the light-curve tests but used in the joint fit and the pixel analysis. A blind search over 1–40 d ranks the signal first; the second peak is its own half-period alias. Both TOI-5997 b and the candidate are shallowest in Sectors 52–53 (233 ± 52 and 246 ± 81 ppm, against 337 ± 29 and 330 ± 39 ppm in all sectors), again suggesting a sector-dependent effect.

The candidate has $R_p = 1.27 \pm 0.09 R_\oplus$ and $b = 0.33 \pm 0.20$, and the free fit gives $\rho_\star = 2.7^{+1.3}_{-1.9} \text{ g cm}^{-3}$, 0.2σ from the stellar value. TOI-5997 b, fitted jointly, has $R_p = 1.44 \pm 0.11 R_\oplus$, in agreement with Collier et al. (2026). The pixel analysis places the source 0.5σ from TOI-5997. As for TOI-678, the transit is detected in the difference images at only 2.7σ after calibration, so no neighbour is excluded by both tests. The nearest, TIC 1508332922 ($T = 17.1$, $28''$), is 4.1σ from the fitted position by the bootstrap test but only 2.5σ by the calibrated χ^2 . For TOI-5997 b, which is detected at a similar level but needs a smaller correction for correlated pixel noise, all four neighbours are excluded (4.2σ calibrated detection).

The ‘Alopeke and PHARO images rule out companions brighter than $\Delta m \approx 7$ beyond $0.5''$ and $\Delta m_{832} \approx 4.6$ beyond $0.1''$, so an undetected companion beyond $0.1''$ would add at most about 1.5% of the light and change the radius by less than 1%. The star’s Hipparcos–Gaia proper-motion anomaly is not significant (S/N of 2.1; Zhang et al. 2024). TRICERATOPS gives FPP = $(3.7 \pm 0.2) \times 10^{-4}$ and NFPP = $(3.4 \pm 0.3) \times 10^{-4}$; for TOI-5997 b the same configuration gives 5.1×10^{-4} . Both values for the candidate are below the thresholds commonly used for statistical validation (Sect. 6).

6. DISCUSSION

6.1. How strong are the candidates?

Each signal is a SPOC detection that we recover with consistent depths in seasons separated by up to six years, in SAP as well as PDCSAP flux, without odd–even differences, secondary eclipses or aliases, and from a position consistent with the target. Two lie in systems that already host a planet or candidate, which raises their prior probability (Lissauer et al. 2012). We do not call them validated, for three reasons. Their S/N is modest: 8–10 with the conservative noise estimate. All three pass the S/N threshold of 10 that we set beforehand, but only with the time-averaging β . The TOI-678 candidate rests on three transits, and our own blind search would not have found it. And the usual TRICERATOPS validation thresholds, FPP < 0.015 and NFPP < 10^{-3} (Giacalone et al. 2021), are met only by the TOI-5997 candidate (Table 5). We still list that one as a candidate: its S/N is the lowest of the three, and TRICERATOPS assumes the signal is astrophysical, so it does not test for an instrumental false alarm. One more season of TESS or CHEOPS photometry at the predicted times would settle this. For HIP 31126 the host is unknown, which leaves the candidate’s radius uncertain by a factor of 1.27 and its insolation by a factor of 2.5.

6.2. Observations that would validate them

Table B1 lists the next transits.

HIP 31126. The pair is $3.3''$ apart, so seeing-limited photometry with point-spread-function fitting can measure A and B separately. A transit on B would be about 2.8 ppt deep in B’s light ($V \approx 12.7$) and could be detected or ruled out with a 1–2-m telescope in one transit. A transit on A would be about 0.75 ppt deep in A’s light ($V \approx 10.7$) and last 2.1 h; detecting it needs better precision, a larger telescope or several transits. Either way the host would be identified. Imaging closer to the star than the SOAR limits ($\Delta I = 2.5$ at $0.15''$) would also reduce the unresolved-companion scenarios. The expected radial-velocity semi-amplitude, about 1.3 m s^{-1} if the host is A, is within reach of ESPRESSO or NIRPS.

TOI-678. The transit is 0.9 ppt deep and 6.1 h long on a $V = 11.6$ star near the southern ecliptic pole, out of reach of CHEOPS, and TESS is not scheduled to observe it in its ninth year (to 2027 September). Seeing-limited photometry of the field during a transit would test the neighbours directly: they would need eclipses of 1–90% of their own light (1.2% for TIC 294395934, which dominates the NFPP), easy to detect even with partial coverage. Detecting the transit on TOI-678 itself needs good conditions or a larger telescope. The next transits are on 2026 December 29 and 2027 May 8 (Table B1).

TOI-5997. TESS will observe TOI-5997 again in Sectors 117 and 118 (2027 May 10 to July 3; positions from tess-point, Burke et al. 2020), which should cover three transits (2027 May 24, June 7 and June 21) unless they fall in data gaps. The 0.3-ppt transit is too shallow for most ground-based telescopes but within reach of CHEOPS. Seeing-limited photometry of the neighbours during one transit would remove the remaining neighbour scenarios, and with a 14.2-d period there are many opportunities each season. The radial-velocity signal, about 0.7 m s^{-1} , is small.

6.3. The candidates in context

Figure 3 places the candidates among the 4 740 transiting planets in the Planetary Systems Composite Parameters table of the NASA Exoplanet Archive (DOI 10.26133/NEA13, retrieved 2026 October 8; Akesson et al. 2013; Christiansen et al. 2025). Only 5 of them have periods longer than 100 d, radii of $2\text{--}4 R_\oplus$ and Sun-like hosts brighter than $V = 12$; the TOI-678 candidate ($V = 11.6$, $T_{\text{eq}} \approx 391 \text{ K}$) would add to this small group. The HIP 31126 candidate would be a small planet in a probably bound pair of M dwarfs at 24 pc; if it orbits B it receives 1.2 times the Earth’s insolation, and 2.9 times if it orbits A. TOI-5997 would host two small planets with a period ratio of 2.514, 0.6% from the 5:2 commensurability; transit-timing variations, if present, would be small.

7. SUMMARY

1. Three periodic signals detected by the TESS pipeline, none of them a TOI, have consistent depths across transits and seasons, no odd–even difference or secondary eclipse, no detectable alias and the same depth in SAP flux. Their S/N is 8.4–9.8 with the conservative (full-phase) noise estimate and 10.8–11.1 with the time-averaging estimate.
2. Around TOI-678, the 130.145-d signal is a $3.60 \pm 0.28 R_\oplus$ candidate receiving $3.9 S_\oplus$, the second candidate in the system. It rests on three transits; a fourth, in full-frame data, is inconclusive. Its FPP is 0.021 ± 0.002 .

TABLE 5
FALSE-POSITIVE PROBABILITIES FROM TRICERATOPS.

	TOI-678 (130.1 d)	HIP 31126 (29.29 d)	TOI-5997 (14.22 d)
<i>Adopted: archival imaging; mean and standard deviation of 5, 5 and 4 runs</i>			
FPP	0.021 ± 0.002	0.40 ± 0.02	$(3.7 \pm 0.2) \times 10^{-4}$
NFPP	0.016 ± 0.002	0.40 ± 0.02	$(3.4 \pm 0.3) \times 10^{-4}$
Planet on the target (TP + PTP + DTP)	0.979 ± 0.002	0.60 ± 0.02	0.99963 ± 0.00002
Planet on an unresolved companion (STP)	0.0033 ± 0.0013	0.0031 ± 0.0003	$(2.8 \pm 3.0) \times 10^{-5}$
Planet on a resolved neighbour (NTP)	0.0050 ± 0.0008	0.38 ± 0.02 (on B)	$(5.5 \pm 1.2) \times 10^{-6}$
All eclipsing-binary scenarios	0.012 ± 0.002	0.0174 ± 0.0018	$(3.3 \pm 0.3) \times 10^{-4}$
<i>Imaging and pixel-level exclusions (one run for HIP 31126)</i>			
FPP	as adopted	0.41	as adopted
NFPP	as adopted	0.40	as adopted
<i>TESS only: no imaging, no pixel-level exclusions (one run, 2×10^5 draws)</i>			
FPP	0.025	0.63	0.051

Scenario names follow [Giacalone et al. \(2021\)](#). No neighbour of TOI-678 or TOI-5997 is excluded by both pixel-level tests, so for them this configuration is the adopted one. With the adopted configuration, TOI-678.01 has FPP = 0.0039 and TOI-5997 b 5.1×10^{-4} .

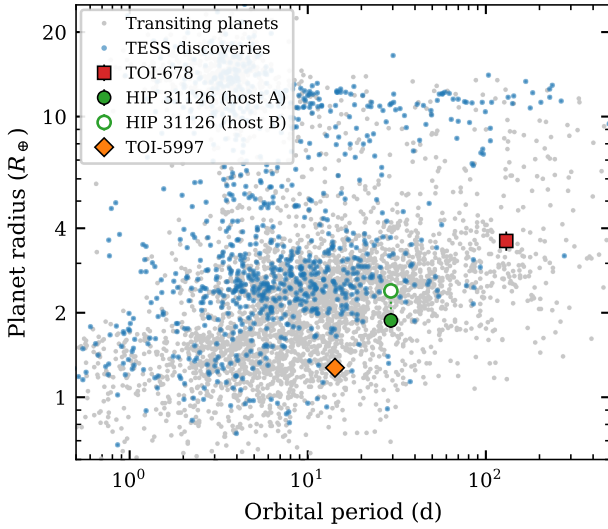


FIG. 3.— Periods and radii of transiting planets in the NASA Exoplanet Archive (grey; TESS discoveries in blue) and of the three candidates. The two possible radii of the HIP 31126 candidate are joined by a dotted line.

3. Around HIP 31126 (GJ 237), the 29.286-d signal is a planet candidate of $1.88 \pm 0.11 R_{\oplus}$ if it orbits A or $2.39 \pm 0.14 R_{\oplus}$ if it orbits B. Eclipsing binaries carry 0.0174 ± 0.0018 of the probability; the host is not known.
4. Around TOI-5997, the 14.216-d signal is a $1.27 \pm 0.09 R_{\oplus}$ candidate beside the validated TOI-5997 b. Its FPP $((3.7 \pm 0.2) \times 10^{-4})$ and NFPP $((3.4 \pm 0.3) \times 10^{-4})$ are below the usual validation thresholds; we keep it as a candidate until more data confirm the modest-S/N signal.
5. A PRF fit to TESS difference images places each signal within about 1σ of its target and places the 14.8-d CTOI in the HIP 31126 field on a neighbour, most likely an eclipsing binary. For the two signals weakest in the difference images (TOI-678 and TOI-5997) it does not exclude neighbours at 3σ once correlated pixel noise is calibrated.
6. Seeing-limited photometry during one transit would identify the host of the HIP 31126 candidate and test the neighbours of TOI-678 and TOI-5997.

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Use of generative AI. The analysis code, the literature checks and the draft text were prepared with the assistance of Claude (Anthropic), a large language model, under the author’s direction, and an AI-assisted review of the draft was carried out. Most derived numbers in the text and tables are

written by a script from the analysis outputs. The author checked the results and takes responsibility for the content.

DATA AVAILABILITY

All data used here are public at MAST, ExoFOP and the NASA Exoplanet Archive. The code, the input files

with checksums, posterior samples of the transit fits and the TRICERATOPS outputs are provided as ancillary files with the arXiv version of this paper. The full package, which also contains the light curves and pixel response models extracted from MAST, will be archived at Zenodo.

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APPENDIX

A. CONTROL SIGNALS

Table A1 applies the same tests to four other signals in the same light curves. TOI-678.01 and TOI-5997 b pass them except for the consistency of their depths: those of TOI-678.01 vary between transits and between seasons by more than the formal errors ($p = 0.014$ and 0.015), and those of TOI-5997 b between seasons ($p = 0.049$). In both stars the candidate is also shallower in the same sectors (Sects. 5.1 and 5.3), which points to an instrumental effect common to those sectors rather than to a property of either signal. The 14.8-d signal near HIP 31126 comes from a neighbour and is most likely an eclipsing binary (Sect. 5.2). The 6.56-d TCE of TOI-678 has consistent depths but falls below our S/N threshold; it may deserve attention as more data accumulate.

TABLE A1

CONTROL SIGNALS, ANALYSED WITH THE SAME CODE AS THE CANDIDATES.

	TOI-678.01	TOI-5997 b	14.8-d signal near HIP 31126	6.56-d TCE of TOI-678
Period (d)	11.330	5.655	14.810	6.562
Transits used	32	20	11	50
Box depth (ppm)	852 ± 37	337 ± 29	1413 ± 70	248 ± 35
S/N: full-phase β ; time-averaging β	23.2; 31.7	11.6; 12.0	20.2; 27.3	7.2; 8.7
Depth at random phases (σ)	19.0	11.3	18.1	7.5
Per-transit p ; per-season p	0.014; 0.015	0.51; 0.049	0.86; 0.68	0.87; 0.69
Odd-even difference (σ)	2.0	0.7	2.4	1.2
Target from fitted pixel position: Mahalanobis; χ^2 (σ)	1.1; 1.6	0.6; 0.7	> 10 ; > 10	–
Neighbours able to cause the signal; excluded	9; 7	4; 4	5; 4	–
FPP (adopted configuration)	0.0039	5.1×10^{-4}	–	–

For the 14.8-d signal, A and B are among the excluded stars; its source is TIC 293689273. The pixel analysis of the controls uses the target pixel files processed for the candidates (for TOI-678.01, Sectors 8, 61 and 90 only; for the 14.8-d signal, not Sectors 87 and 98). The 6.56-d TCE was not localised.

B. UPCOMING TRANSITS

Table B1 lists the next predicted transits of the three candidates.

TABLE B1

PREDICTED TRANSITS AFTER 2026 OCTOBER 15 (FOR TOI-5997, FROM THE START OF ITS 2027 OBSERVING SEASON).

Star	Mid-transit (UTC)	BJD _{TDB}	σ (min)	T_{14} (h)
TOI-678	2026-12-29 08:31	2461403.8560	16	6.1
	2027-05-08 12:01	2461534.0015	17	6.1
	2027-09-15 15:30	2461664.1470	18	6.1
HIP 31126	2026-10-17 04:51	2461330.7026	8	2.1
	2026-11-15 11:42	2461359.9882	8	2.1
	2026-12-14 18:33	2461389.2738	8	2.1
	2027-01-13 01:24	2461418.5594	8	2.1
	2027-02-11 08:16	2461447.8450	8	2.1
TOI-5997	2027-04-11 08:32	2461506.8566	10	3.2
	2027-04-25 13:43	2461521.0726	10	3.2
	2027-05-09 18:55	2461535.2887	10	3.2
	2027-05-24 00:06	2461549.5047	10	3.2

From the fits of Table 4, with timing uncertainties multiplied by the full-phase β . The UTC times are barycentric; the light-travel time to an observer on Earth differs by up to ± 8 min. Whether a transit is observable depends on the site.

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